

Topologically trivial states induced by strong spin-orbit coupling and Chern insulators in doped $X(\text{C}_{21}\text{N}_3\text{H}_{15})$ ($X = \text{Ta}, \text{Hf}$) metal-organic frameworks

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We study topological properties of two-dimensional (2D) bi-triangular lattices with multiple p and d orbitals. Tight-binding (TB) model calculations show that the multiple p (p_x/p_y) and d ($d_{xy}/d_{x^2-y^2}$ or d_{xz}/d_{yz}) bands in the 2D bi-triangular lattices, with C_3 symmetry, are all degenerate at the Γ point with quadratic non-Dirac band dispersions and gapped at the K/K' points. Spin-orbit coupling (SOC) is found not only resulting in the topologically nontrivial states, but also the topologically trivial states. In the 2D bi-triangular lattices, a topological phase transition from a topologically nontrivial state to a trivial state can be triggered by increasing the atomic SOC strength. Our first-principles calculations predict a large band-gap quantum anomalous Hall state (about 30 meV) and topological phase transitions in a 2D metal-organic frameworks $X(\text{C}_{21}\text{N}_3\text{H}_{15})$ ($X = \text{Ta}, \text{Hf}$), rationalized well by the TB model built above. Our results are helpful to deeply understand the topological states and provide theoretical guidance for the search of 2D topological materials based on the multiple p and d orbitals.

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I. INTRODUCTION

Both quantum spin Hall (QSH) and quantum anomalous Hall (QAH) states have attracted considerable attention in condensed matter physics and material science due to their topologically protected dissipationless edge states [1–7], which may greatly promote the development of low-power-consumption spintronics and electronics devices. For the QSH effect, two notable models, Kane-Mele [8] and Bernevig-Hughes-Zhang (BHZ) [9,10] models, were proposed, which provide guidance thoughts to carry out this topological effect in realistic materials. The Kane-Mele model, the first theoretically proposed model for QSH states, is based on a two-dimensional (2D) honeycomb lattice with p_z orbital, in which a finite spin-orbit coupling (SOC) can open band gaps at K and K' points in the Brillouin zone (BZ) and result in a QSH state. In the BHZ model, the QSH state is realized due to the SOC induced band inversion mechanism. If the time-reversal symmetry (TRS) of a QSH state is broken by an appropriate internal exchange field, the quantum anomalous Hall (QAH) state may be realized. So far, the QSH and QAH states have been theoretically predicted to occur in numerous material systems [8,11–32]. However, the QSH state has only been observed in CdTe/HgTe/CdTe and InAs/GaSb/AlSb quantum wells [10,33], and monolayer $1T'$ WTe₂ [34] systems. The QAH state has been realized merely in magnetically doped

(Bi, Sb)₂Te₃ films [35–38]. The reasons can be ascribed to the realization difficulty of the theoretically proposed QSH and QAH materials in current experiments. Therefore, predicting large band-gap QSH and QAH states in new species of materials that can be easily fabricated in experiments is of great significance, which may drastically promote the development of this field.

Recently, theoretical studies demonstrate that multiple p or d orbitals accommodated on an atomic lattice with a C_3 symmetry exhibit a degenerate energy point with quadratic non-Dirac band dispersions at the Γ point [32,39]. When the atomic SOC is taken into account, a band gap will be opened and result in a topologically nontrivial state [32,39] around the Γ point. This proposed mechanism for the topologically nontrivial states in 2D systems is different from the well-known Kane-Mele and BHZ models. In addition, if the multiple p or d orbitals are accommodated on a triangular lattice (C_6 symmetry), the two p bands or two d bands will be simultaneously degenerate at the Γ and K/K' points in the momentum space with quadratic non-Dirac and linear Dirac band dispersions, respectively [40]. The atomic SOC can simultaneously open topologically nontrivial band gaps around the K/K' points [8] and the Γ point [32]. The SOC induced global band gap in the 2D system is, however, a topologically trivial state rather than a nontrivial state due to the coupling effect between the topological states at the Γ and K/K' points [40]. Wang *et al.* demonstrated that a minimal basis of three orbitals (s , p_x , p_y) is required to produce a QSH phase via nearest-neighbor hopping in a 2D triangular lattice [31].

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In this work we systematically investigate the topological properties of 2D bi-triangular lattices, containing two nonequivalent simple triangular lattices, by using tight-binding (TB) models and density functional theory (DFT) calculations. Multiple p (such as p_x and p_y) and d (such as d_{xy} and $d_{x^2-y^2}$) orbitals are considered at the triangular A and B sublattices, respectively. These p and d bands are both found being degenerate at the Γ point with quadratic non-Dirac band dispersions and gapped at the K/K' points before the SOC is turned on. When the atomic SOC is taken into account, a topologically nontrivial (QSH or QAH) band gap is opened at the Γ point, whose magnitude increases with the increase of the atomic SOC. A different trend is, however, obtained for the band gaps at the K/K' points. With the increase of the SOC, the original band gap opened at the K/K' points is found decreasing, closing, and then reopening. A phase transition from a topologically nontrivial state to a trivial state is triggered by the strong SOC interaction in the process, which is counterintuitive since the SOC interactions generally produce the topological nontrivial states in previous reports. The above proposed mechanism can rationalize well

the topological behaviors in the 2D monolayer metal-organic frameworks $X(\text{C}_{21}\text{N}_3\text{H}_{15})$ (TPyB- X) where $X = \text{Ta}$ or Hf . First-principles calculations show that the topologically trivial state in TPyB- X can be tuned to the QAH state by decreasing the SOC strength. Our findings are helpful to deeply understand the topological states and guide the searching for or designing of the topologically nontrivial states based on real materials.

II. RESULTS AND DISCUSSION

A. Topological properties of the p_x/p_y and $d_{xy}/d_{x^2-y^2}$ bands in 2D bi-triangular lattices

The left inset in Fig. 1(a) shows a schematic drawing of a 2D bi-triangular lattice considered, which is formed by two simple triangular lattices, labeled as A and B, respectively. We now study the electronic states of this bi-triangular lattice with p_x and p_y orbitals at the A sites and d_{xy} and $d_{x^2-y^2}$ orbitals at the B sites. The total TB model Hamiltonian of this bi-triangular lattice in the basis of $(p_A, p_B, d_{Bxy}, d_{Bx^2-y^2})$ can be written as

$$H(k) = H_{\text{hop}}(k) + H_{\text{soc}} - H_M = \begin{bmatrix} H_{\text{hop}\uparrow} & 0 \\ 0 & H_{\text{hop}\downarrow} \end{bmatrix} + \begin{bmatrix} H_{\text{soc}\uparrow} & 0 \\ 0 & H_{\text{soc}\downarrow} \end{bmatrix} - \begin{bmatrix} M_A \mathbf{I} & 0 & 0 \\ 0 & M_B \mathbf{I} & 0 \\ 0 & -M_A \mathbf{I} & 0 \\ 0 & 0 & -M_B \mathbf{I} \end{bmatrix}, \quad (1)$$

where H_{hop} , H_{soc} , and H_M represent the hopping term, the on-site SOC term, and the magnetic exchange field term, respectively. The up and down arrows denote the spin-up and spin-down channels, respectively. M_A and M_B are the applied exchange field strengths at the A and B sites, respectively. $\mathbf{I}$ in Eq. (1) is a unit matrix.

The hopping terms including the nearest-neighbor (NN) and next-nearest-neighbor (NNN) interactions for both the spin-up and spin-down channels can be expressed as

$$H_{\text{hop}\uparrow/\downarrow} = \begin{bmatrix} h_{11} & h_{12} & h_{13} & h_{14} \\ h_{21} & h_{22} & h_{23} & h_{24} \\ h_{31} & h_{32} & h_{33} & h_{34} \\ h_{41} & h_{42} & h_{43} & h_{44} \end{bmatrix} \quad (2)$$

in which

$$\begin{aligned} h_{11} &= \varepsilon_p + 2t_{pp\pi} \cos(\sqrt{3}k_y) + \left(\frac{3}{2}t_{pp\sigma} + \frac{1}{2}t_{pp\pi}\right) \left[\cos\left(\frac{3}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) + \cos\left(\frac{3}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) \right], \\ h_{12} &= \frac{\sqrt{3}}{2}(t_{pp\pi} - t_{pp\sigma}) \left[\cos\left(\frac{3}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) - \cos\left(\frac{3}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) \right], \\ h_{21} &= h_{12}, \quad h_{22} = \varepsilon_p + 2t_{pp\sigma} \cos(\sqrt{3}k_y) + \left(\frac{1}{2}t_{pp\sigma} + \frac{3}{2}t_{pp\pi}\right) \left[\cos\left(\frac{3}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) + \cos\left(\frac{3}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) \right], \\ h_{13} &= \left(\frac{3}{8}t_{pd\sigma} + \frac{\sqrt{3}}{4}t_{pd\pi}\right) (e^{i(-\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y)} - e^{i(-\frac{1}{2}k_x - \frac{\sqrt{3}}{2}k_y)}), \\ h_{14} &= \frac{\sqrt{3}}{2}t_{pd\sigma} e^{ik_x} + \left(\frac{\sqrt{3}}{8}t_{pd\sigma} - \frac{3}{4}t_{pd\pi}\right) (e^{i(-\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y)} + e^{i(-\frac{1}{2}k_x - \frac{\sqrt{3}}{2}k_y)}), \\ h_{23} &= t_{pd\pi} e^{ik_x} + \left(-\frac{3\sqrt{3}}{8}t_{pd\sigma} + \frac{1}{4}t_{pd\pi}\right) (e^{i(-\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y)} + e^{i(-\frac{1}{2}k_x - \frac{\sqrt{3}}{2}k_y)}), \end{aligned}$$

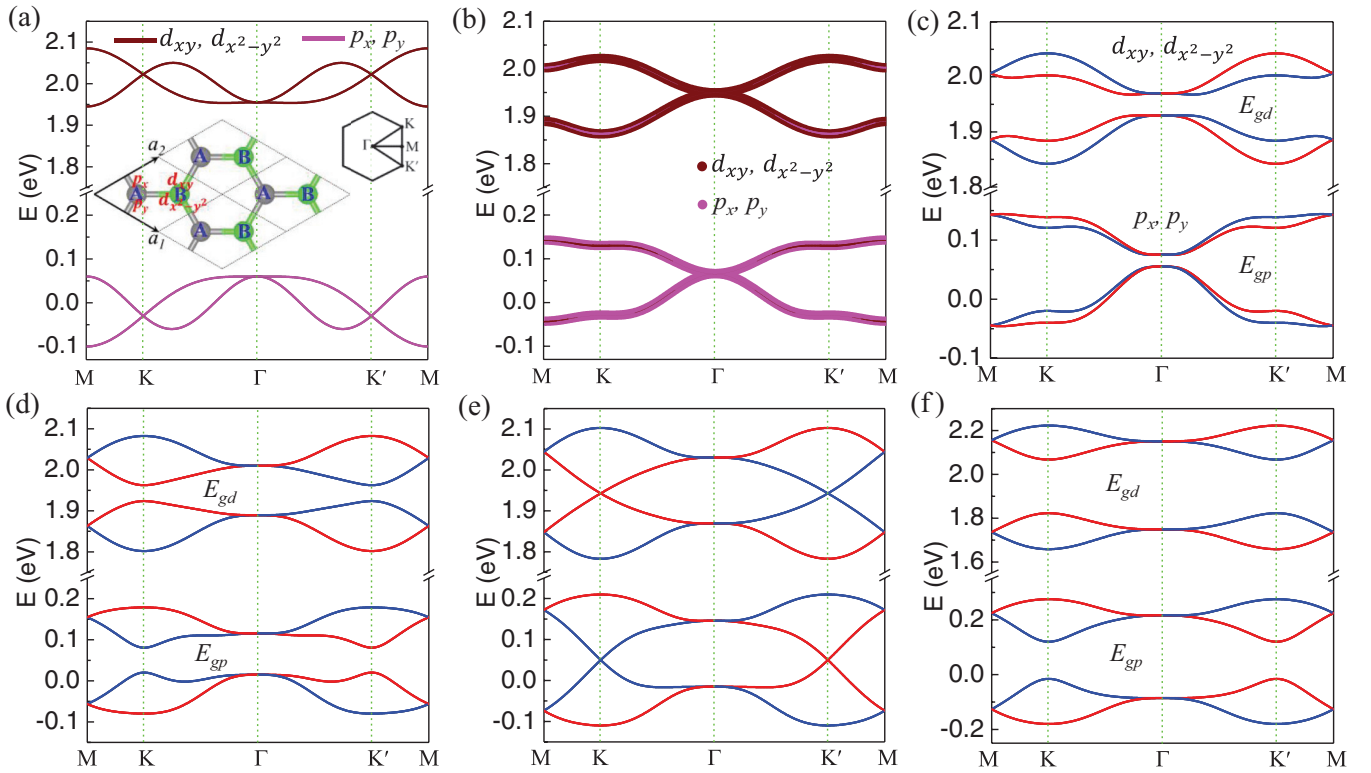


FIG. 1. (a) The calculated TB band structure with the parameters: $\varepsilon_p = 0.0$ eV, $t_{pp\sigma} = 0.03$ eV, $t_{pp\pi} = -0.01$ eV, $M_A = 0.0$ eV, $\lambda_A = 0.0$ eV, $\varepsilon_d = 2.00$ eV, $t_{dd\delta} = -0.01$ eV, $t_{dd\sigma} = -0.03$ eV, $t_{dd\pi} = 0.01$ eV, $M_B = 0.0$ eV, $\lambda_B = 0.0$ eV, $t_{pd\sigma} = 0.0$ eV, and $t_{pd\pi} = 0.0$ eV. Left inset: A schematic drawing of the 2D bi-triangular lattice formed by two nonequivalent triangular lattices, which are labeled as A and B sublattices, respectively. One triangular lattice is with p_x and p_y orbitals and the other is with d_{xy} and $d_{x^2-y^2}$ orbitals. $\vec{a}_1 = \sqrt{3}(\sqrt{3}/2, -1/2)$ and $\vec{a}_2 = \sqrt{3}(\sqrt{3}/2, 1/2)$ are lattice vectors. Right inset: The first BZ and the special k points. (b) The same as (a), except for $t_{pd\sigma} = 0.25$ eV, $t_{pd\pi} = 0.15$ eV. The contributions from the p_x/p_y (magenta dots) and $d_{xy}/d_{x^2-y^2}$ (wine dots) orbitals to the bands are denoted. (c)–(f) The TB band structures with the projection of spin operator $\hat{s}_z$. The red and blue colors indicate the spin-up and spin-down states, respectively. The adopted TB parameters are the same as (b) except for $\lambda_A = 0.01$ eV, $\lambda_B = 0.01$ eV (c), $\lambda_A = 0.05$ eV, $\lambda_B = 0.03$ eV (d), $\lambda_A = 0.08$ eV, $\lambda_B = 0.04$ eV (e), and $\lambda_A = 0.15$ eV, $\lambda_B = 0.10$ eV (f), respectively. E_{gp} and E_{gd} denote the minimum band gaps in the multiple p and d bands after including the SOC, respectively.

$$\begin{aligned}
 h_{24} &= \left(-\frac{3}{8}t_{pd\sigma} - \frac{\sqrt{3}}{4}t_{pd\pi} \right) \left(e^{i(-\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y)} - e^{i(-\frac{1}{2}k_x - \frac{\sqrt{3}}{2}k_y)} \right), \\
 h_{33} &= \varepsilon_d + 2t_{dd\pi} \cos(\sqrt{3}k_y) + \frac{1}{8}(9t_{dd\sigma} + 4t_{dd\pi} + 3t_{dd\delta}) \left[\cos\left(\frac{3}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) + \cos\left(\frac{3}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) \right], \\
 h_{34} &= -\frac{\sqrt{3}}{8}(3t_{dd\sigma} - 4t_{dd\pi} + t_{dd\delta}) \left[\cos\left(\frac{3}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) - \cos\left(\frac{3}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) \right], \\
 h_{44} &= \varepsilon_d + \frac{1}{2}(3t_{dd\sigma} + t_{dd\delta}) \cos(\sqrt{3}k_y) + \frac{1}{8}(3t_{dd\sigma} + 12t_{dd\pi} + t_{dd\delta}) \left[\cos\left(\frac{3}{2}k_x - \frac{\sqrt{3}}{2}k_y\right) + \cos\left(\frac{3}{2}k_x + \frac{\sqrt{3}}{2}k_y\right) \right], \\
 h_{31} &= -h_{13}^*, h_{32} = -h_{23}^*, \quad h_{41} = -h_{14}^*, \quad h_{42} = -h_{24}^*, \quad h_{43} = h_{34}.
 \end{aligned} \tag{3}$$

In Eq. (3) ε_p and ε_d are the on-site energies for the p_x/p_y and $d_{xy}/d_{x^2-y^2}$ orbitals, respectively. $t_{pp\sigma}$, $t_{pp\pi}$, $t_{dd\delta}$, $t_{dd\sigma}$, and $t_{dd\pi}$ are the NNN hopping parameters. $t_{pd\sigma}$ and $t_{pd\pi}$ are the NN hopping parameters, expressing the hopping interactions between the A and B sublattices.

The on-site SOC term of this bi-triangular lattice does not include the coupling between the different spin components,

which means that the spin-up and spin-down channels of this model Hamiltonian are separated from each other. For each spin channel, the on-site SOC can be written as a 4×4 matrix

$$H_{\text{soc}\uparrow/\downarrow} = s \cdot \begin{bmatrix} 0 & -\lambda_A i & 0 & 0 \\ \lambda_A i & 0 & 0 & 0 \\ 0 & 0 & 0 & 2\lambda_B i \\ 0 & 0 & -2\lambda_B i & 0 \end{bmatrix}, \tag{4}$$

where λ_A and λ_B are the atomic SOC strengths of the A and B sites, respectively. $s = +1$ and -1 represent the spin-up and spin-down channels, respectively.

The calculated band structures of the bi-triangular lattice based on the above constructed TB model Hamiltonian are given in Fig. 1. The calculated band structure without considering the NN hopping is formed by the bands of the two isolated triangular lattices with multiple p (p_x and p_y) and d (d_{xy} and $d_{x^2-y^2}$) orbitals, respectively, as shown in Fig. 1(a) with following TB parameters: $\varepsilon_p = 0.0$ eV, $t_{pp\sigma} = 0.03$ eV, $t_{pp\pi} = -0.01$ eV, $M_A = 0.0$ eV, $\lambda_A = 0.0$ eV, $\varepsilon_d = 2.00$ eV, $t_{dd\delta} = -0.01$ eV, $t_{dd\sigma} = -0.03$ eV, $t_{dd\pi} = 0.01$ eV, $M_B = 0.0$ eV, $\lambda_B = 0.0$ eV, $t_{pd\sigma} = 0.0$ eV, and $t_{pd\pi} = 0.0$ eV. The multiple p (p_x and p_y) and d (d_{xy} and $d_{x^2-y^2}$) bands in the two separated triangular lattices are both simultaneously degenerate at the Γ and K/K' points with quadratic non-Dirac and linear Dirac band dispersions, respectively [Fig. 1(a)]. When the NN hopping interactions between the two triangular lattices are taken into account, the band structure of the real bi-triangular lattice will be obtained, as shown in Fig. 1(b), with $t_{pd\sigma} = 0.25$ eV, $t_{pd\pi} = 0.15$ eV. Figure S1 in the Supplemental Material [41] and Fig. 1(b) give the obtained orbital-projected band structures, from which we can observe that the bands present a mixture of the p and d orbitals to a certain extent due to the NN hopping interactions between the two sublattices. Obviously the degeneracy at the K and K' points is eliminated while the degeneracy of the p (p_x and p_y) and d (d_{xy} and $d_{x^2-y^2}$) bands at the Γ point keeps [Fig. 1(b)]. According to the previous studies [32,40], in the simple triangular lattice with only one set of sublattice, the SOC induced local band gaps around the K/K' and Γ points are all topologically nontrivial. The global band gap in this simple triangular lattice is, however, topologically trivial rather than nontrivial due to the coupling effect between the local topological states at the K/K' and Γ points [see Fig. 1(a)] [40]. By adding the NN hopping interaction between the two triangular sublattices, topological trivial band gaps are opened at the K/K' points [Fig. 1(b), without SOC]. In the following, the interesting band structures and topological properties of this bi-triangular lattice with respect to the SOC strength are explored systematically.

Based on Fig. 1(b), the band evolution with different SOC strengths are illustrated in Figs. 1(c) to 1(f). The SOC strengths at the A and B sublattices adopted in Fig. 1(c) are $\lambda_A = 0.01$ eV and $\lambda_B = 0.01$ eV. The atomic SOC induces band gaps for both the p and d orbitals around the Γ point [Fig. 1(c)]. Additionally, the spin splitting appears in the bands due to the inversion symmetry breaking in this system. The different colors in Fig. 1(c) express the projection of the spin operator $\hat{s}_z$, i.e., $\langle \psi_{n\mathbf{k}} | \hat{s}_z | \psi_{n\mathbf{k}} \rangle$. The red and blue colors indicate the spin-up and spin-down states, respectively. For the p orbitals in Fig. 1(c), it is interesting to note that the band gap of the two spin-down bands at the K point is slightly smaller than that at the K' point (blue curves), while the trend is opposite for the spin-up bands. For the d orbitals in Fig. 1(c), the whole tendency is opposite to that of the p orbitals. The SOC strengths are increased from Figs. 1(c) to 1(f). Explicitly, with the increase of the SOC strength, the band gaps around the Γ point in the p_x/p_y (or $d_{xy}/d_{x^2-y^2}$) bands increase monotonously, while the band gap

at the K/K' points first decreases to zero and then reopens and keeps increasing, implying possible topological phase transitions.

To quantitatively describe the band evolution, the band gaps at the K/K' points without the SOC are denoted as $\delta_{K/K'}$. The band gap changes at the K/K' points due to the SOC are denoted by $\Delta_{K/K'}$, defined as $\Delta_{K/K'} = \delta_{K/K'} - \eta_{K/K'}$, where $\eta_{K/K'}$ is the energy gap at the K/K' points with SOC. Note that $\eta_{K/K'}$ is defined as a negative value if the band gap is reopened at the K/K' points after it is closed with the increase of the SOC strength. The E_{gp} (E_{gd}) denotes the minimum band gap opened between the two p (d) bands after including the SOC. When $\Delta_{K/K'}(p) (\approx 2\lambda_A) = \delta_{K/K'}(p)$, the band gaps in the p bands are closed at the K and K' points for both of the spin-down and spin-up channels, as shown in Fig. 1(e). Similarly, for the d orbitals, when $\Delta_{K/K'}(d) (\approx 4\lambda_B) = \delta_{K/K'}(d)$, the band gaps in the d bands are closed at the K and K' points for both of the spin-up and spin-down channels.

The Berry curvatures and Chern numbers are calculated to identify the topological properties of the SOC-induced band gaps in the above bi-triangular lattice with multiple p (p_x/p_y) and d ($d_{xy}/d_{x^2-y^2}$) orbitals. The Berry curvatures $\Omega(\mathbf{k})$ are calculated by [42,43]

$$\Omega(\mathbf{k}) = \sum_n f_n \Omega_n(\mathbf{k}),$$

$$\Omega_n(\mathbf{k}) = -2\text{Im} \sum_{m \neq n} \frac{\hbar^2 \langle \psi_{n\mathbf{k}} | v_x | \psi_{m\mathbf{k}} \rangle \langle \psi_{m\mathbf{k}} | v_y | \psi_{n\mathbf{k}} \rangle}{(E_m - E_n)^2}, \quad (5)$$

where the summation is over all of the occupied states, E_n is the eigenvalue of $|\psi_{n\mathbf{k}}\rangle$, v_{xy} are the velocity operators, and f_n is the Fermi-Dirac distribution function. The Chern number C can be obtained by integrating the Berry curvatures $\Omega(\mathbf{k})$ over the first BZ, as $C = \frac{1}{2\pi} \sum_n \int_{\text{BZ}} d^2k \Omega_n$. Since the spin-up and spin-down subspaces of the constructed TB Hamiltonian are decoupled, one can calculate the Berry curvatures for the spin-up and spin-down subspaces independently.

Figure 2(a) shows the spin-up bands (red curves) of the p orbitals obtained from the spin-up subspace of the TB Hamiltonian with the parameters adopted in Fig. 1(c). The violet dots in Fig. 2(a) display the obtained Berry curvatures when the Fermi level (E_F) is located inside the SOC-induced band gap ($E_{gp\uparrow}$). In Fig. 2(a), the obtained $\Omega(\mathbf{k})$ around the Γ point is positive and with an approximate M shape, while the $\Omega(\mathbf{k})$ around the K and K' points have the opposite signs. Integrating the $\Omega(\mathbf{k})$ in Fig. 2(a) around the Γ and K/K' points, the numbers of $C_{p\Gamma\uparrow} = +1$ and $C_{pK\uparrow} + C_{pK'\uparrow} = 0$ are obtained. Therefore, the total Chern number for the spin-up subspace of the TB Hamiltonian when the E_F is located inside the SOC-induced band gap ($E_{gp\uparrow}$) is $C_{p\uparrow} = C_{p\Gamma\uparrow} + C_{pK\uparrow} + C_{pK'\uparrow} = C_{p\Gamma\uparrow} = +1$. Figure 2(b) presents the corresponding spin-down bands (blue curves) of the p orbitals and Berry curvatures (violet dots). The $\Omega(\mathbf{k})$ around the Γ point for the spin-down p_x/p_y bands are negative and with an approximate W shape, while the $\Omega(\mathbf{k})$ around the K and K' points also have the opposite signs. Therefore, the obtained total Chern number for the spin-down p_x/p_y bands with the E_F located inside the SOC-induced gap ($E_{gp\downarrow}$) is $C_{p\downarrow} = -1$.

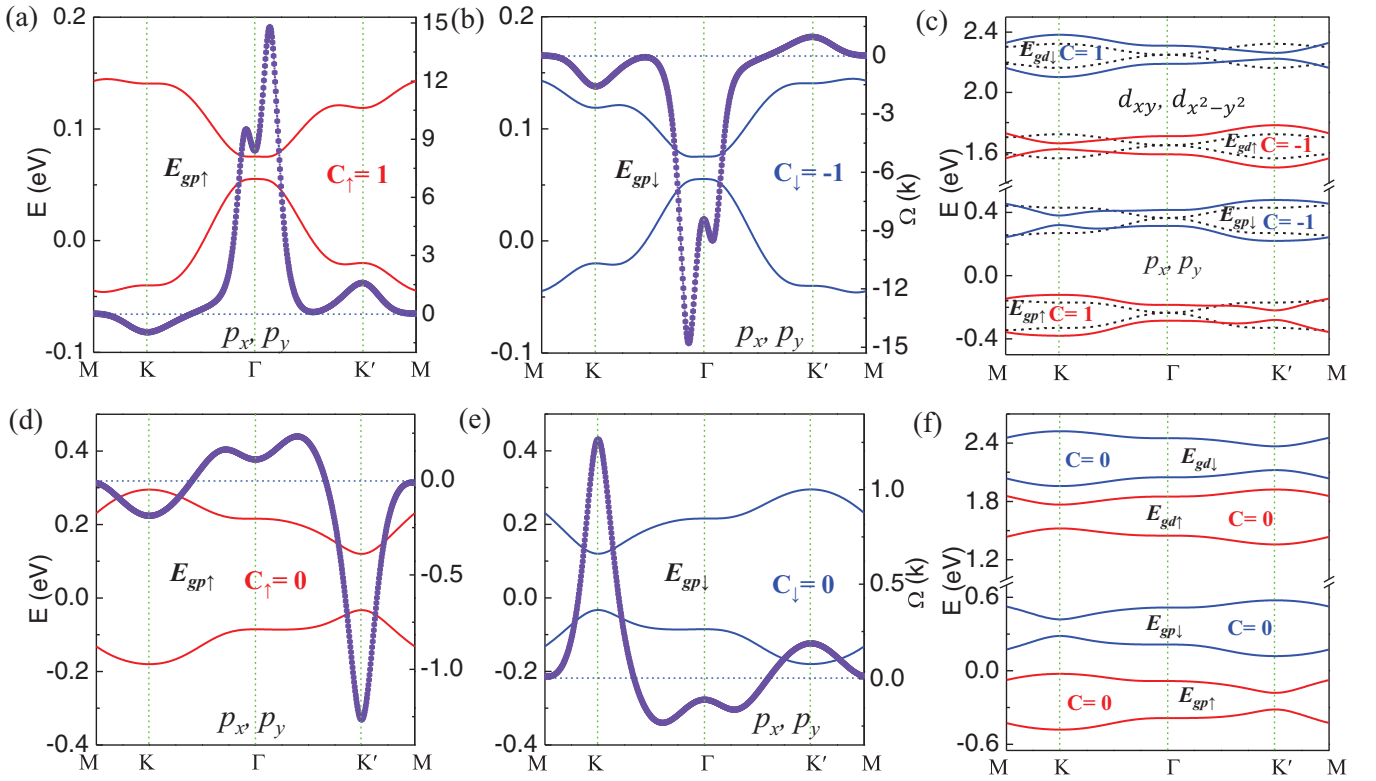


FIG. 2. (a) and (b) The calculated Berry curvatures (violet dots) of the spin-up and spin-down bands of the p (p_x and p_y) orbitals in Fig. 1(c), respectively. The corresponding bands (red and blue curves) are also displayed. (c) Magnetized band structures with the projection of spin operator $\hat{s}_z$, calculated based on Fig. 1(d) with additional $M_A = M_B = 0.30$ eV. The black dashed curves are the corresponding bands without SOC ($\lambda_A = \lambda_B = 0.0$ eV). The red and blue colors indicate the spin-up and spin-down states, respectively. (d)–(f) The same as (a)–(c), respectively, except for based on Fig. 1(f).

Hence the obtained spin Chern number for the p orbitals in Fig. 1(c) is $C_{ps} = \frac{1}{2}(C_{p\uparrow} - C_{p\downarrow}) = +1$, indicating the SOC-induced global band gap (E_{gp}) in the two p bands in Fig. 1(c) can display QSH effect. Thus, the appearing mechanisms of topological phases of the p orbitals in this work are different from those in Ref. [31], where the s - p band inversion is necessary to realize the topological state. The topological behaviors of the spin-up and spin-down bands of the d ($d_{xy}/d_{x^2-y^2}$) orbitals in Fig. 1(c) can be analyzed analogously. The obtained results are given in Figs. S2(a) and S2(b). The calculated spin Chern number $C_{ds} = \frac{1}{2}(C_{d\uparrow} - C_{d\downarrow}) = -1$ for the band gap of the two d ($d_{xy}/d_{x^2-y^2}$) orbitals also proves the QSH state existing within the d bands. Figure 2(c) show the calculated TB band structures of the bi-triangular lattice with the same TB parameters as those adopted in Fig. 1(d) except for the exchange interactions considered with $M_A = M_B = 0.30$ eV, to be discussed.

Figure 2(d) show the spin-up bands (red curves) of the p orbitals and Berry curvatures (violet dots) obtained from the TB Hamiltonian with the parameters adopted in Fig. 1(f). One can observe that the $\Omega(k)$ around the Γ and K points in Fig. 2(d) have the same sign as those in Fig. 2(a), while the $\Omega(k)$ around K' point become negative in Fig. 2(d), opposite to that in Fig. 2(a), due to the band gap in the spin-up p orbitals first closed and then reopened at the K' point from Figs. 1(c) to 1(f). Integrating the $\Omega(k)$ in Fig. 2(d), the Chern number of $C_{p\uparrow} = 0$ is obtained. Figure 2(e) displays the corresponding spin-down bands (blue curves) of the p (p_x/p_y) orbitals and

Berry curvatures (violet dots). For the spin-down p bands, the band gap first closes and then reopens at the K point rather than the K' point from Figs. 1(c) to 1(f), resulting in the sign change of the $\Omega(k)$ around the K point in Fig. 2(e), compared to Fig. 2(b). Thus, integrating the $\Omega(k)$ in Fig. 2(e), the Chern number of $C_{p\downarrow} = 0$ is also obtained. Therefore, the obtained spin Chern number for the band gap of the p orbitals in Fig. 1(f) is $C_{ps} = \frac{1}{2}(C_{p\uparrow} - C_{p\downarrow}) = 0$, displaying the SOC-induced band gap of E_{gp} in Fig. 1(f) is topologically trivial rather than the topologically nontrivial. The Berry curvatures of the spin-up and spin-down bands of the d ($d_{xy}/d_{x^2-y^2}$) orbitals in Fig. 1(f) are given in Figs. S2(c) and S2(d). The calculated spin Chern number $C_{ds} = \frac{1}{2}(C_{d\uparrow} - C_{d\downarrow}) = 0$ for the d bands indicates the band gap between the two d orbitals in Fig. 1(f) is topological trivial, similar to the case of the p orbitals in the same figure.

Figures 2(c) and 2(f) show the calculated TB band structures of the constructed bi-triangular lattice with the same TB parameters as those adopted in Figs. 1(d) and 1(f), respectively, except for spin polarization considered with $M_A = M_B = 0.30$ eV. The spin-up and spin-down bands in Figs. 2(c) and 2(f) are entirely separated from each other, as shown in the projection of spin operator $\hat{s}_z$. The dotted curves in Fig. 2(c) plot the bands without the SOC. When the SOC is turned on, band gaps appear around the Γ point for all four pairs of the bands. For the spin-up p bands and spin-down d bands, the band gaps increase at the K point and decrease at the K' point, compared to the cases without SOC, while for the spin-down p

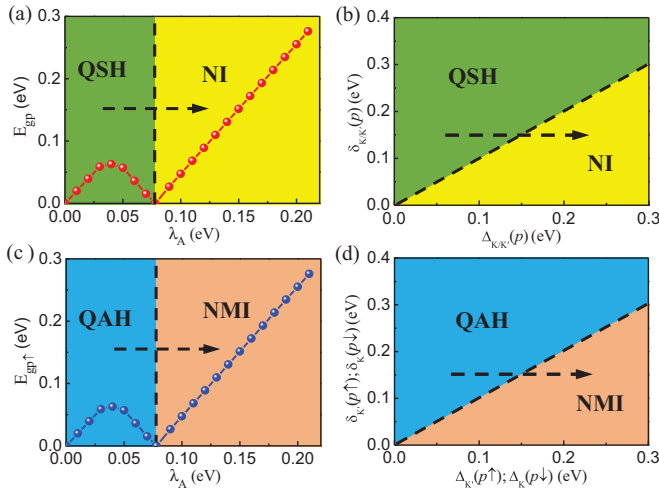


FIG. 3. (a) Topological phase diagram with respect to the SOC strength. The red dotted curve shows the energy band gap (E_{gp}) for the p_x and p_y bands as a function of the SOC strength (λ_A). The other TB parameters are the same as in Fig. 1(b). (b) Topological phase diagram of the multiple p bands with C_3 symmetry in the bi-triangular lattices. $\delta_{K/K'}(p)$ represents the band gap at the K/K' point in the multiple p bands before the SOC is included. $\Delta_{K/K'}(p)$ represents the SOC induced band-gap change at the K/K' point in the multiple p bands. (c) Topological phase diagram with respect to the SOC strength (λ_A) in the spin polarized system. The blue dotted curve shows the band gap ($E_{gp\uparrow}$) for the spin-up p_x and p_y bands as a function of the SOC strength (λ_A). The other TB parameters are the same as in Fig. 2(c). (d) Topological phase diagram of the band gap in the two fully spin-polarized p bands in the bi-triangular lattice after including the SOC. $\delta_{K'}(p\uparrow)$ represents the band gap at the K' point in the two spin-up p bands before the SOC is included. $\Delta_{K'}(p\uparrow)$ represents the SOC induced band gap change at the K' point in the two spin-up p bands. $\delta_K(p\downarrow)$ and $\Delta_K(p\downarrow)$ are the same as $\delta_{K'}(p\uparrow)$ and $\Delta_{K'}(p\uparrow)$, respectively, except for the band gaps at the K point in the two spin-down p bands.

bands and spin-up d bands, the trends are opposite, as shown in Fig. 2(c). The four global band gaps ($E_{gp\uparrow}$, $E_{gp\downarrow}$, $E_{gd\uparrow}$, and $E_{gd\downarrow}$) in Fig. 2(c) with SOC can all present QAH effects, identified by the calculated Chern numbers $C = \pm 1$ when the E_F are located inside the band gaps of the p or d spin-up and spin-down bands. In Fig. 2(f), the Chern numbers $C = 0$ are obtained for all the four band gaps, indicating the SOC induced four global band gaps ($E_{gp\uparrow}$, $E_{gp\downarrow}$, $E_{gd\uparrow}$, and $E_{gd\downarrow}$) in Fig. 2(f) all present the behavior of a normal magnetic insulator (NMI) rather than a QAH state. Our calculations show that the topological properties of the multiple p and d bands (with C_3 symmetry) in the 2D bi-triangular lattice are not influenced by the relative energy positions of the p and d bands.

The red dotted curve in Fig. 3(a) shows the band gap E_{gp} in the multiple p (p_x and p_y) bands as a function of the SOC strength λ_A with the other TB parameters the same as in Fig. 1(b). With the increase of the SOC strength, the band gap (E_{gp}) first increases to a largest value and then decreases to zero. If the SOC strength continues increasing, the E_{gp} will be reopened and keeps increasing. According to above discussion, a topological phase transition from a QSH state to a nor-

mal insulator (NI) will occur when the E_{gp} decreases to zero in the process of the increasing of the SOC strength, as shown in Fig. 3(a). This result is counterintuitive since it is the SOC interactions that generally produce the topological nontrivial states reported previously. A strong SOC is usually desired much to open a large topologically nontrivial band gap [16]. Our obtained result, however, indicates that a strong SOC may not always be positive to topological states. It can cause the disappearance of the topological state. The topological phase transition can also be expressed by the band gap changes, as illustrated in Fig. 3(b). When the absolute values of the SOC induced band gap changes $\Delta_{K/K'}(p)$ for the p bands at the K (or K') points are smaller than the trivial band gaps $\delta_{K/K'}(p)$ for the p bands opened at the K (or K') points due to the hopping interactions between the nonequivalent A and B triangular sublattices, that is $\Delta_{K/K'}(p) < \delta_{K/K'}(p)$, the QSH state will be achieved. Otherwise, $\Delta_{K/K'}(p) > \delta_{K/K'}(p)$, the NI will be obtained [Fig. 3(b)]. If the NN hopping parameters between the two nonequivalent triangular sublattices in the TB model are increased, the band gaps $\delta_{K/K'}(p)$ will be increased, which will make the topological nontrivial state more robust against the SOC strength, as illustrated in Fig. 3(b). Similar behaviors as in Figs. 3(a) and 3(b) also survive for the d ($d_{xy}/d_{x^2-y^2}$) bands in the bi-triangular lattice system studied.

When the exchange interactions are included in the TB model, the topological phase transition from a QAH state to a NMI will occur with the increase of the SOC [Figs. 3(c) and 3(d)]. The blue dotted curve in Fig. 3(c) shows the spin-up band gap of the p bands ($E_{gp\uparrow}$) as a function of the SOC strength (λ_A) with the other TB parameters adopted the same as in Fig. 2(c). Different from the Fig. 3(b), the spin-up and spin-down bands are completely separate in Fig. 3(d). The phase transition depends on the spin index, as indicated in Fig. 3(d).

B. Topological properties of the d_{xz}/d_{yz} bands in 2D bi-triangular lattices

In the constructed bi-triangular lattice, if the A sites are set to be p_x and p_y orbitals and the B sites are set to be d_{xz} and d_{yz} orbitals, there are no interactions between the two nonequivalent triangular lattices and the band gaps cannot be opened at the K/K' points before the SOC is included. To investigate the topological properties of the d_{xz} and d_{yz} orbitals in the bi-triangular lattice with C_3 symmetry, we design another bi-triangular lattice with p_z orbitals at the A sites and d_{xz}/d_{yz} orbitals at the B sites, as shown in the inset of Fig. 4(a). The TB model Hamiltonian of this bi-triangular lattice is given in Supplemental Note 1 in the Supplemental Material [41]. Figure 4(a) shows the calculated band structure of this constructed bi-triangular lattice without considering the NN hopping, the TB parameters adopted in Fig. 4(a) are $\varepsilon_p = 0.0$ eV, $t_{pp\pi} = -0.01$ eV, $M_A = 0.0$ eV, $\varepsilon_d = 3.00$ eV, $t_{dd\delta} = 0.02$ eV, $t_{dd\pi} = -0.03$ eV, $M_B = 0.0$ eV, $\lambda_B = 0.0$ eV, and $t_{pd\pi} = 0.0$ eV. The d_{xz}/d_{yz} bands in the triangular lattice (with C_6 symmetry) are also simultaneously degenerate at the Γ and K/K' points with quadratic non-Dirac and linear Dirac band dispersions, respectively, as shown in Fig. 4(a). When the NN hopping interactions ($t_{pd\pi} = 0.25$ eV) is included, the band gaps opened at the K and K' points and the bands

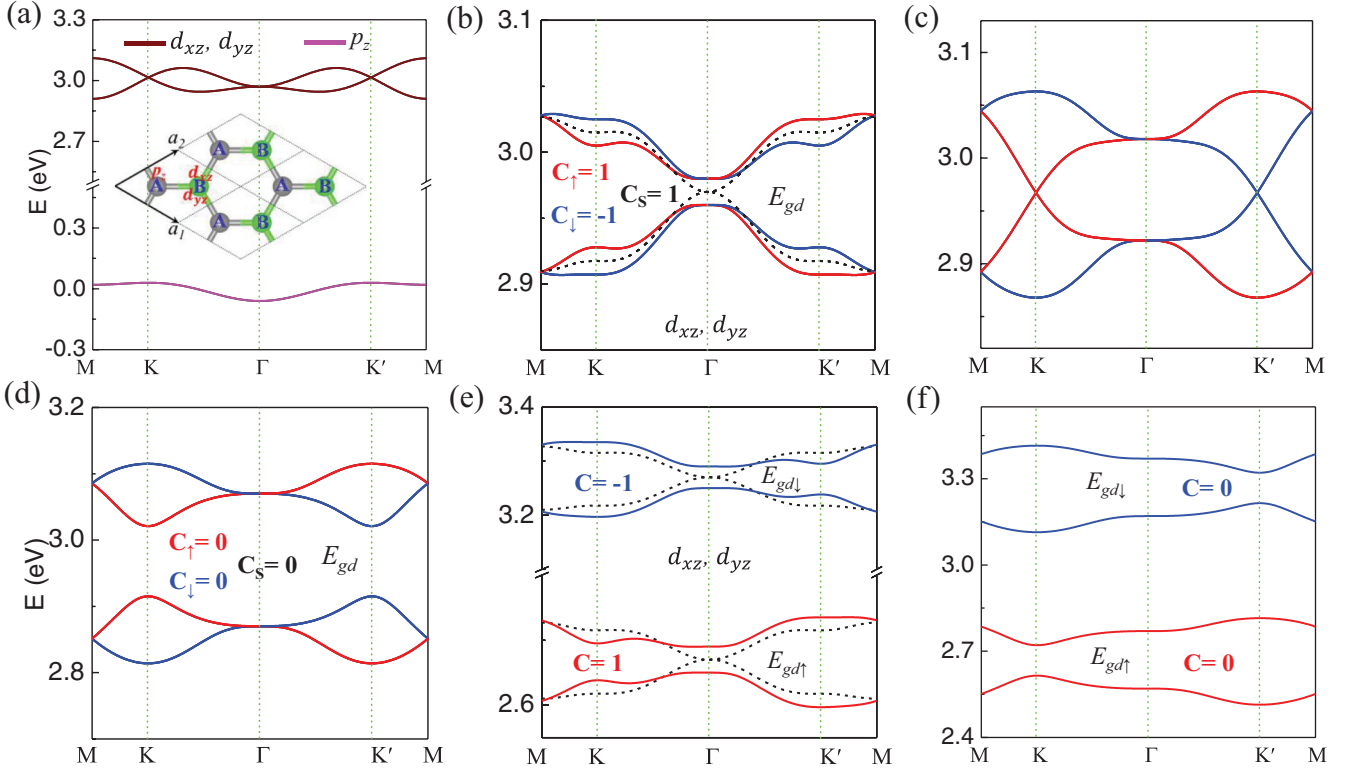


FIG. 4. (a) The calculated TB band structure with the parameters: $\varepsilon_p = 0.0$ eV, $t_{pp\pi} = -0.01$ eV, $M_A = 0.0$ eV, $\varepsilon_d = 3.00$ eV, $t_{dd\delta} = 0.02$ eV, $t_{dd\pi} = -0.03$ eV, $M_B = 0.0$ eV, $\lambda_B = 0.0$ eV, and $t_{pd\pi} = 0.0$ eV. Inset: A schematic drawing of the 2D bi-triangular lattice. One triangular lattice (A sublattice) is with p_z orbital and the other (B sublattice) is with d_{xz} and d_{yz} orbitals. $\vec{a}_1 = \sqrt{3}(\sqrt{3}/2, -1/2)$ and $\vec{a}_2 = \sqrt{3}(\sqrt{3}/2, 1/2)$ are lattice vectors. (b) Black dashed curves: The parameters are the same as those of the d bands in (a), except for $t_{pd\pi} = 0.25$ eV. Colored curves: the TB band structures with the projection of spin operator $\hat{s}_z$, in which the SOC interaction ($\lambda_B = 0.01$ eV) is considered. The red and blue colors indicate the spin-up and spin-down states, respectively. (c) and (d) The same as the colored curves in (b) except for $\lambda_B = 0.048$ eV (c) and $\lambda_B = 0.10$ eV (d), respectively. (e) and (f) Band structures with the projection of spin operator $\hat{s}_z$. The bands are calculated with the same TB parameters as adopted in colored curves in (b), except for $M_B = 0.30$ eV, $\lambda_B = 0.02$ eV (e) and $M_B = 0.30$ eV, $\lambda_B = 0.10$ eV (f), respectively. The black dashed curves in (e) are the corresponding bands without SOC. The red and blue curves in (e) and (f) indicate the spin-up and spin-down states, respectively.

(with C_3 symmetry) are only degenerate at the Γ point with quadratic non-Dirac band dispersions, as shown by the black dashed curves in Fig. 4(b). The evolution of these d (d_{xz}/d_{yz}) bands [black dashed curves in Fig. 4(b)] with respect to the SOC strength is displayed in Figs. 4(b) to 4(d), where the SOC strengths adopted are (b) $\lambda_B = 0.01$ eV, (c) $\lambda_B = 0.048$ eV, and (d) $\lambda_B = 0.10$ eV. With the increase of the SOC strength, the energy gap in the d_{xz}/d_{yz} bands at the Γ point increases monotonously, while the energy gaps at the K and K' points first decrease to zero and then reopen and keep increasing. These trends are similar to the cases in Fig. 1. Especially, when $\Delta_{K/K'}(d) (\approx 2\lambda_B) = \delta_{K/K'}(d)$, the band gaps in the multiple d (d_{xz}/d_{yz}) bands are closed at the K and K' points with linear Dirac band dispersions which are respectively formed by the spin-up and spin-down states, as shown in Fig. 4(c).

From the calculated Berry curvatures [violet dots in Figs. S3(a) and S3(b)], the obtained Chern numbers for the spin-up and spin-down subspaces of the TB Hamiltonian with the parameters adopted in Fig. 4(b) when the E_F is located inside the band gap in the d_{xz}/d_{yz} bands is $C_{d\uparrow} = +1$, and $C_{d\downarrow} = -1$, respectively. Hence the obtained spin Chern number in Fig. 4(b) in the d_{xz}/d_{yz} bands is $C_{ds} = \frac{1}{2}(C_{d\uparrow} - C_{d\downarrow}) = +1$, indicating the SOC-induced band gap (E_{gd}) in Fig. 4(b)

is the QSH state. Dissimilarly, in Fig. 4(d), the obtained Chern numbers for the spin-up and spin-down subspaces of the TB Hamiltonian are $C_{d\uparrow} = 0$ and $C_{d\downarrow} = 0$, respectively. Hence, the obtained spin Chern numbers in Fig. 4(d) when the E_F is located inside the band gap in the d_{xz}/d_{yz} bands is $C_{ds} = \frac{1}{2}(C_{d\uparrow} - C_{d\downarrow}) = 0$, indicating the SOC-induced band gap (E_{gd}) in Fig. 4(d) presents behaviors of a NI rather than the QSH effect. Although the feature of the band evolution of the d_{xz}/d_{yz} orbitals with the increase of the SOC is the similar to that of the $d_{xy}/d_{x^2-y^2}$ orbitals, the distribution of the $\Omega(\mathbf{k})$ of the d_{xz}/d_{yz} bands is opposite to that of the $d_{xy}/d_{x^2-y^2}$ bands, as shown in Figs. S2 and S3. Thus, the spin Chern number obtained for d_{xz}/d_{yz} bands ($C_s = +1$) is opposite to that obtained for $d_{xy}/d_{x^2-y^2}$ bands ($C_s = -1$) due to the different orbital behaviors.

Figures 4(e) and 4(f) show the calculated band structures with the TB parameters the same as adopted in the colored curves in Fig. 4(b), except for two following parameters of $M_B = 0.30$ eV, $\lambda_B = 0.02$ eV and $M_B = 0.30$ eV, $\lambda_B = 0.10$ eV, respectively. The SOC-induced two global band gaps ($E_{gd\uparrow}$ and $E_{gd\downarrow}$) in Fig. 4(e) can both present QAH effects. The two SOC induced global band gaps ($E_{gd\uparrow}$ and $E_{gd\downarrow}$) in Fig. 4(f) are both a topologically trivial state. Therefore,

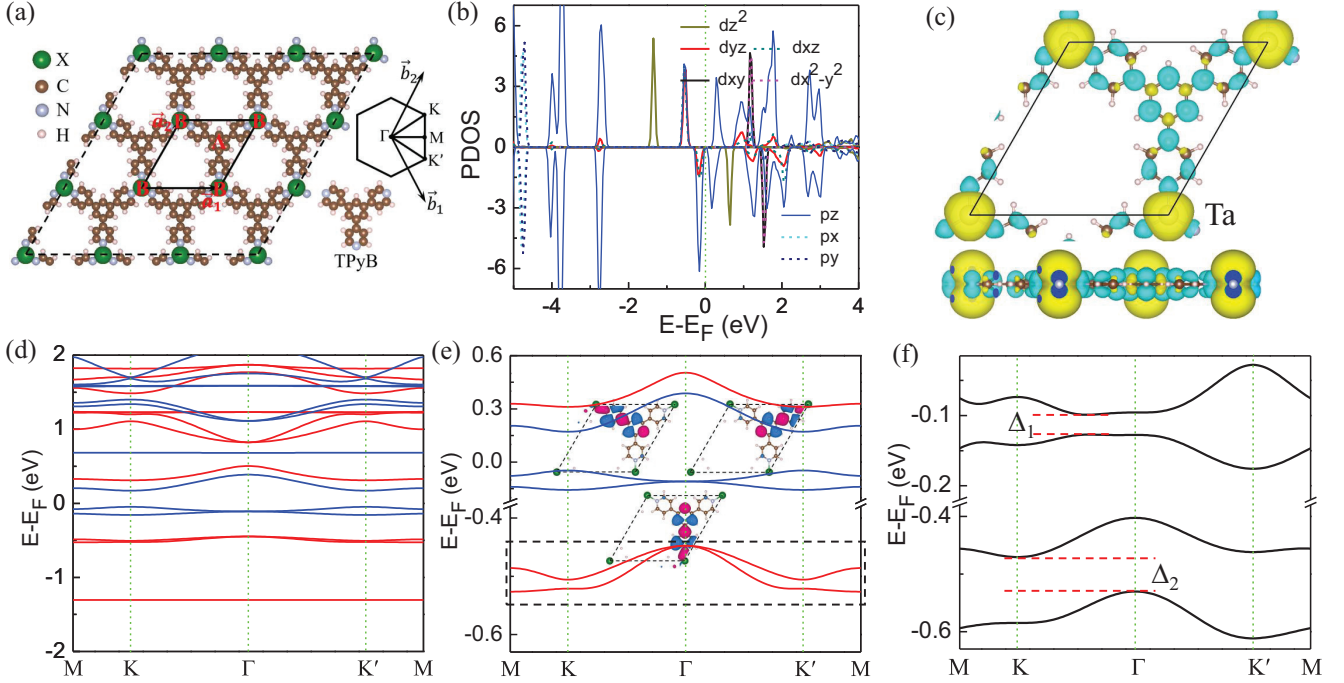


FIG. 5. (a) Top view of the 2D $X(\text{C}_{21}\text{N}_3\text{H}_{15})$ (TPyB- X) monolayer with lattice vectors $\vec{a}_1$ and $\vec{a}_2$. The solid lines indicate the unit cell. Insets: The TPyB ($\text{C}_{21}\text{N}_3\text{H}_{15}$) molecule and the first BZ with the high symmetry k points. The red labels A and B in the unit cell represent the A and B sublattices in the bi-triangular lattice, respectively. (b) The calculated partial densities of states of TPyB-Ta. The positive and negative values of PDOSs express the spin-up and spin-down states, respectively. (c) The spatial distribution of spin-polarized electron density for TPyB-Ta. The yellow and light blue colors denote the net spin-up and spin-down charge densities, respectively. (d) The calculated spin-polarized band structure of TPyB-Ta. The red and blue curves indicate the spin-up and spin-down bands, respectively. (e) Magnification of (d) around the E_F . Inset: The obtained three MLWFs of the three spin-down bands around the E_F in (e). (f) The corresponding magnified band structure of TPyB-Ta with SOC.

similar to the case in Figs. 2(c) and 2(f), a topological phase transition from a QAH state to a NMI will occur in the spin-polarized d_{xz}/d_{yz} bands with the increase of the SOC strength.

According to above discussion, we can observe that the atomic SOC can induce topologically trivial as well as nontrivial states in the bi-triangular lattice with multiorbital (p_x/p_y , $d_{xy}/d_{x^2-y^2}$, and d_{xz}/d_{yz} with C_3 symmetry). One cannot always expect large SOC interactions to produce the topologically nontrivial states with large band gaps since strong SOC interactions can also lead to the breakdown of the topological state. And the topologically trivial states can be induced by strong SOC for various orbitals. In the following, the electronic structures and topological properties of 2D triangular lattice metal-organic frameworks (equivalent to the 2D bi-triangular lattice) $X(\text{C}_{21}\text{N}_3\text{H}_{15})$ (TPyB- X) where X represents 5d heavy transition metal elements Ta and Hf are systematically investigated.

C. SOC-induced topologically trivial states and Chern insulators in $X(\text{C}_{21}\text{N}_3\text{H}_{15})$ ($X=\text{Ta}, \text{Hf}$)

The geometry and electronic structures of the 2D metal-organic framework $X(\text{C}_{21}\text{N}_3\text{H}_{15})$ ($X=\text{Ta}, \text{Hf}$) are calculated by using the projector augmented wave (PAW) [44] formalism based on density functional theory, as implemented in the Vienna *ab initio* simulation package (VASP) [45]. The Perdew-Burke-Ernzerhof generalized-gradient approximation

(GGA-PBE) is used to describe the exchange and correlation functional [46]. To take into account the correlation effects of the Ta (Hf) 5d electrons, the GGA+ U method [47] is adopted, and the on-site Coulomb interaction U and exchange interaction J are set to be 2.0 and 1.0 eV, respectively. A vacuum space of larger than 15 Å is used to avoid the interaction between adjacent slabs. The plane-wave cutoff energy is set to be 450 eV. The convergence criterion for the total energy is set to be 10^{-6} eV. The Monkhorst-Pack k -point grids of $10 \times 10 \times 1$ are adopted. All atoms in the unit cell are allowed to relax until the Hellmann-Feynman force on each atom is smaller than 0.01 eV/Å. The maximally localized Wannier functions (MLWFs) [48,49] of the TPyB- X ($X=\text{Ta}, \text{Hf}$) are constructed by using the Wannier90 package [50]. The numerical algorithm described in Ref. [51] is employed to calculate the Berry curvatures.

The 2D monolayer metal-organic frameworks [26–30] are a category of very important material systems that are proposed to host the topologically nontrivial states due to their special lattice structures which are formed by metal atoms and organic molecules. For example, the QSH (or QAH) effects were reported in metal-organic frameworks with honeycomb lattices [26,27] or kagome lattices [28,29]. Figure 5(a) displays the atomic structure of the monolayer TPyB- X lattice, whose unit cell contains one X atom and one TPyB ($\text{C}_{21}\text{N}_3\text{H}_{15}$) molecule. Since the TPyB molecule can be considered effectively as a superatom, the TPyB- X lattice can be regarded as the bi-triangular lattice discussed above,

with the TPyB molecule located at the A site and the X atom located at the B site. Thus, the two nonequivalent triangular lattices in this bi-triangular lattice are formed by the TPyB molecules and the X atoms, respectively. Among this material family, $\text{Fe}(\text{C}_{21}\text{N}_3\text{H}_{15})$ has been successfully synthesized in experiments [52]. Here the electronic structures and topological properties of TPyB-X ($X=\text{Hf}$, Ta) are investigated. The adopted 5d heavy elements ($X=\text{Ta}$, and Hf) in TPyB-X can give rise to relatively strong SOC to examine the TB model results obtained. The structural stability of the TPyB-Ta and TPyB-Hf has been examined by *ab initio* molecular dynamics (AIMD) and phonon calculations; details are discussed in Supplemental Note 2 of the Supplemental Material [41]. All atoms in TPyB-Ta and TPyB-Hf monolayers are coplanar after structure relaxation, as shown in Fig. 5(a). The optimized lattice constants $|\vec{a}_1| = |\vec{a}_2| = a$ of TPyB-Ta and TPyB-Hf are 13.61 and 13.71 Å, respectively.

Figure 5(b) displays the calculated atomic orbital projected densities of states (PDOSs) of the TPyB-Ta. Figure 5(c) shows the spatial distribution of spin-polarized electron for TPyB-Ta, from which we find that the magnetic moment of TPyB-Ta is contributed by Ta atoms and C/N atoms in the system. The approximate three spin-up occupied d_{z^2} , d_{xz} , and d_{yz} orbitals and the two spin-down occupied molecular orbitals just below the E_F give rise to a 1.0 μ_B net magnetic moment for the TPyB-Ta in one unit cell. Our first-principles calculations show that the ground state of the TPyB-Ta lattice is the ferromagnetic (FM) state, which is more stable than the nonmagnetic state by about 472 meV/unit cell. Figure 5(d) shows the calculated spin-polarized band structure of TPyB-Ta along the high symmetry lines in the first BZ [as illustrated in the inset of Fig. 5(a)]. The TPyB-Ta is a direct semiconductor with a band gap of about 218 meV when the SOC is not included. Figure 5(e) is the magnification of Fig. 5(d) around the E_F , in which three groups of featured spin-up and spin-down bands are displayed, respectively. The three spin-down bands in Fig. 5(e) contain a doublet state and a singlet state, which are mainly contributed by the p_z orbitals of the C and N atoms of the TPyB molecule. The inset in Fig. 5(e) shows the calculated MLWFs for the three spin-down bands by using the Wannier90 package [50]. Interestingly, the three obtained MLWFs exhibit the same features as the sp^2 hybridized orbitals, with a threefold rotational symmetry around the center of the three orbitals. Therefore, the three spin-down bands around the E_F can be regarded as three spin-down molecular orbital bands which are formed by the sp^2 hybrid molecular orbitals. They can be transformed into s , p_x , and p_y molecular orbitals of the TPyB molecule. The two spin-up bands within the dashed rectangular box in Fig. 5(e) primarily come from Ta d_{xz} and d_{yz} orbitals. In Fig. 5(e), the calculated DFT band structure of the TPyB-Ta lattice without SOC shows that the multiple p (p_x/p_y) bands (spin-down states) and the multiple d (d_{xz}/d_{yz}) bands (spin-up states) both are degenerate only at the Γ point with quadratic non-Dirac band dispersions and gapped at the K/K' points, consistent with the TB bands obtained in Figs. 1(b) and 4(b) (dashed curves), respectively. The TPyB-Ta lattice can be regarded as the bi-triangular lattice, formed by the TPyB molecule triangular lattice with p (p_x/p_y and p_z) orbitals at the A sites and the Ta triangular lattice with d ($d_{xy}/d_{x^2-y^2}$ and d_{xz}/d_{yz})

orbitals at the B sites. Its electronic and topological properties can be physically understood by the TB model Hamiltonians of the bi-triangular lattice.

Before the SOC is considered, the trivial band gaps at the K (K') point of the two doublet bands are $\delta_K(p \downarrow) = 109.4$ meV and $\delta_K(d \uparrow) = 15.5$ meV due to the NN hopping interactions between the two nonequivalent triangular lattices [Fig. 5(e)]. Due to the hybridization between the Ta d orbitals and the TPyB molecular orbitals, the SOC strength of the p orbital doublet states are enlarged effectively. When the SOC is taken into account, by comparing the K and K' points of Figs. 5(e) and 5(f), we can observe that the SOC-induced band-gap changes at the K point for the p and d orbital doublet bands are $\Delta_K(p \downarrow) = 40$ meV and $\Delta_K(d \uparrow) = 130$ meV, respectively. The SOC-induced global band gaps in the spin-down p doublet bands and in the spin-up d doublet bands are $\Delta_1 = 28.3$ meV and $\Delta_2 = 59.7$ meV, respectively [Fig. 5(f)]. It is interesting to note that the energy gap of the two p (or d) bands at the K point is smaller than that at the K' point [Fig. 5(f)]. This phenomenon is consistent with the features of the p_x/p_y (or d_{xz}/d_{yz}) bands predicted in the TB model calculations given in Fig. 2(b) [or Fig. 4(f)]. The TB parameters obtained by fitting to the DFT bands of the TPyB-Ta with SOC [Fig. 5(f)] are $\varepsilon_{p_{x/y}} = -0.32$ eV, $\varepsilon_{p_z} = -2.7$ eV, $t_{pp\sigma} = 0.01$ eV, $t_{pp\pi} = -0.001$ eV, $M_A = 0.17$ eV, $\lambda_A = 0.017$ eV, $\varepsilon_{d_{xy/x^2-y^2}} = 1.4$ eV, $\varepsilon_{d_{xz/yz}} = -0.32$ eV, $t_{dd\delta} = -0.001$ eV, $t_{dd\sigma} = -0.01$ eV, $t_{dd\pi} = 0.013$ eV, $M_B = 0.17$ eV, $\lambda_B = 0.065$ eV, $t_{pd\sigma} = 0.225$ eV, and $t_{pd\pi} = 0.086$ eV. The DFT and the obtained TB band structures of the TPyB-Ta are given in Fig. S7. Based on the analysis of Figs. 3(b) and 3(d), we can judge that the SOC-induced gap (Δ_1) in the two spin-down p bands can present QAH effects due to $\Delta_K(p \downarrow) < \delta_K(p \downarrow)$, while the SOC-induced gap (Δ_2) in the two spin-up d bands has behaviors of a NMI due to $\Delta_K(d \uparrow) > \delta_K(d \uparrow)$. The actual topological properties of these two SOC-induced band gaps of Δ_1 and Δ_2 are identified by the calculations of the Berry curvatures, Chern numbers, and edge states.

The bands around the SOC-induced band gap Δ_1 of TPyB-Ta obtained from the Wannier interpolation approach (dashed orange curves) are given in Fig. 6(a), matching perfectly with those obtained from the DFT method (black solid curves). The black dashed curves in Fig. 6(a) show the corresponding bands without SOC. The calculated Berry curvatures $\Omega(\mathbf{k})$ when the E_F is located inside the SOC-induced band gap Δ_1 mainly distribute around the SOC-induced band gap around the Γ point. The distributions of the obtained $\Omega(\mathbf{k})$ along the high symmetry lines around the Γ point are mainly negative and with an approximate W shape, as shown in Fig. 6(a) (blue dotted curve). The $\Omega(\mathbf{k})$ in the 2D momentum space exhibits the C_3 symmetry [Fig. 6(b)]. By integrating the obtained $\Omega(\mathbf{k})$ over the first BZ, the nonzero integer Chern number of $C = -1$ is obtained, indicating the QAH state of the SOC-induced band gap Δ_1 . The calculated anomalous Hall conductivity σ_{xy} as a function of the position of E_F is plotted in Fig. 6(c). The Chern number of $C = -1$ is obtained when the E_F is located inside the band gap Δ_1 , while the Chern number of $C = 0$ is obtained when the E_F is located inside the band gap Δ_2 , displaying the NMI behavior of the band gap Δ_2 . The band

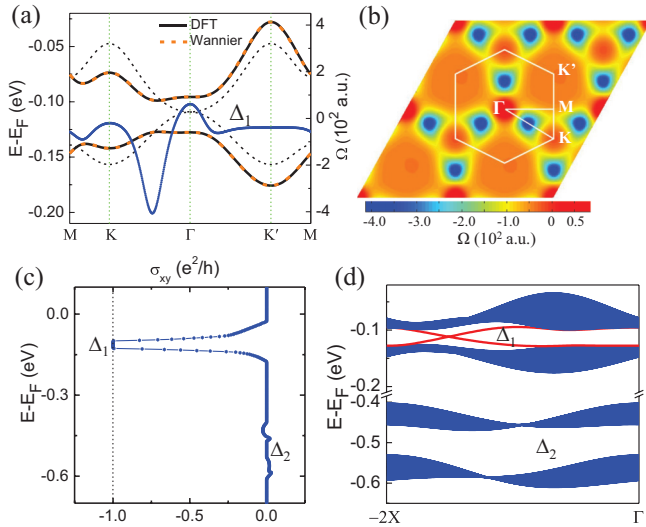


FIG. 6. Topological properties of TPYB-Ta. (a) The magnified bands of TPYB-Ta around the band gap of Δ_1 , obtained by using the DFT method (solid black curves) and Wannier interpolation approach (dashed orange curves). The black dashed curves give the corresponding bands without SOC. The blue dots show the calculated Berry curvatures when the E_F is tuned into the Δ_1 . (b) The corresponding distribution of the Berry curvatures of (a) in 2D momentum space. (c) The calculated anomalous Hall conductivity σ_{xy} as a function of the position of E_F . (d) The calculated bands of a one-dimensional TPYB-Ta ribbon system. The blue curves denote the projected bulk energy bands, while the red curves within the SOC-induced band gap represent the edge states localized at the two edges of the ribbon.

structure of a TPYB-Ta nanoribbon is also calculated by using the TB model Hamiltonian constructed from the MLWFs. As shown in Fig. 6(d), the blue curves denote the projected bulk states and the red curves within the SOC-induced band gap denote the edge states. The two edge states within the band gap Δ_1 represent one chiral conducting channel existing on each side of the nanoribbon sample, confirming the QAH state with $C = -1$. No edge states are obtained within the band gap Δ_2 , confirming the topologically trivial state within this band gap. Therefore, the SOC-induced gaps Δ_1 and Δ_2 are located in the QAH region and NMI region of the phase diagram in Fig. 3(d), respectively. Our first-principles calculations show that the E_F of TPYB-Ta can be tuned into the band gap Δ_1 by doping one hole into a unit cell, with the calculated QAH gap of 26.1 meV (Fig. S8). The corresponding hole doping concentration is about $6.2 \times 10^{13} \text{ cm}^{-2}$, which can be experimentally achieved through currently advanced gating technologies. The carrier density up to $4 \times 10^{14} \text{ cm}^{-2}$ has been realized in experiments [53,54]. The doping concentration needed to tune the E_F into the QAH gap (Δ_1) in this work may be acquired by conventional SiO_2 gating [55], the approach of employing ferroelectric substrates [56], or the method of solid polymer electrolytes [53], etc.

Figure 7(a) shows the magnified bands of TPYB-Ta in the rectangular dashed box in Fig. 5(e). Since the SOC-induced band gap (Δ_2) in these bands is topological trivial, it is meaningful to achieve topological nontrivial states by

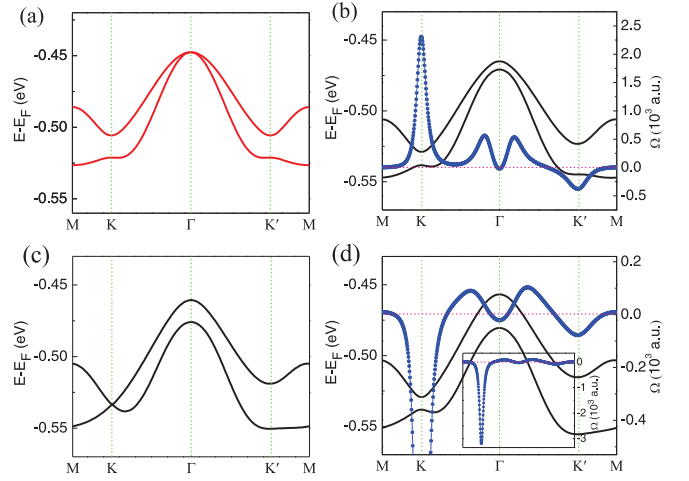


FIG. 7. The magnified bands of TPYB-Ta in the rectangular dashed box in Fig. 5(e). (b)–(d) The calculated band structures of (a) when the SOC strength ($\lambda = x\lambda_0$) is set to be $x = 0.05$ (b), $x = 0.13$ (c), and $x = 0.20$ (d), respectively. The blue dotted curves in (b) and (d) are the calculated Berry curvatures when the E_F is set to be inside the SOC induced local band gaps.

decreasing the SOC strength according to Fig. 3(c) or 3(d). The band evolution of Fig. 7(a) with small SOC strengths are displayed in Figs. 7(b) to 7(d). The adopted SOC strength is defined as $\lambda = x\lambda_0$, where x and λ_0 are the scale factor [0.05 (b), 0.13 (c), 0.20 (d)] and the original SOC strength, respectively. The band gap at the K point in Fig. 7(b) is decreased and the band gap at the K' point is increased, compared to those in Fig. 7(a). The blue dotted curve in Fig. 7(b) shows the calculated Berry curvatures when the curving E_F is set to be located inside the SOC induced band gap. The Chern number $C = 1$ is obtained by integrating the $\Omega(\mathbf{k})$ in the first BZ, indicating this SOC-induced band gap in Fig. 7(b) presents the QAH effect. When the applied SOC strength is increased further to be $\lambda = 0.13\lambda_0$, the band gap at the K point will be closed [Fig. 7(c)]. If the SOC strength continues increasing, the topological phase transition occurs. A Chern number $C = 0$ is obtained from the Berry curvature in Fig. 7(d), indicating the SOC-induced band gap in Fig. 7(d) is topologically trivial. Thus, the topological properties of the multiple $d(d_{xz}/d_{yz})$ bands in the TPYB-Ta can indeed be tuned by the SOC strength, consistent with the topological phase diagram plotted in Fig. 3 obtained from the constructed TB models. The disappearance of topological nontrivial states due to the strong SOC interaction is, therefore, confirmed by the DFT calculations.

The electronic structures and topological properties of the TPYB-X ($X=\text{Hf}$) are also systematically studied. Our calculations show that the FM state of TPYB-Hf is more stable than the nonmagnetic state by about 630 meV/unit cell. The six featured spin-up and spin-down bands of TPYB-Hf in Fig. S9(c) around the E_F can be recognized to be formed by sp^2 hybrid molecular orbitals, which can be equivalently transformed into s , p_x , and p_y molecular orbitals of the TPYB molecule. Therefore, the TPYB-Hf lattice can be treated as the 2D bi-triangular lattice, formed by the TPYB molecule triangular lattice with multiple p (p_x/p_y) orbitals

as the A sublattice and the Hf triangular lattice with multiple d ($d_{xy}/d_{x^2-y^2}$) orbitals as the B sublattice. The two SOC-induced band gaps (Δ_1 and Δ_2) will present QAH effects with the band gaps of $\Delta_1 = 17.8$ meV and $\Delta_2 = 34.4$ meV, respectively. The mechanisms of the two nontrivial band gaps can also be comprehended well with the built TB model. The detail information of the electronic structures and topological properties of TPbB-Hf are given in Supplemental Note 3 in the Supplemental Material [41].

III. CONCLUSIONS

We first systematically investigate the topological properties of the 2D bi-triangular lattices with multiorbital by using the TB model calculations. The TB model Hamiltonians of the bi-triangular lattices contain multiple p (p_x/p_y) and multiple d ($d_{xy}/d_{x^2-y^2}$ or d_{xz}/d_{yz}) orbitals are constructed. The multiple p and d bands in the lattices with C_3 symmetry are both degenerate at the Γ point with quadratic non-Dirac band dispersions and are gapped at the K/K' points due to the interactions between the two nonequivalent triangular lattices. We find the topological phase transitions from topologically nontrivial states to trivial states will occur with the increase of the atomic SOC strength. Namely, the SOC not only can result in the topologically nontrivial (QSH or QAH) states but also the topologically trivial (NI or NMI) states in the

three types of the multiple bands (p_x/p_y , $d_{xy}/d_{x^2-y^2}$, and d_{xz}/d_{yz}) in the bi-triangular lattices. The electronic structures and topological behaviors of monolayer metal-organic frameworks of $X(C_{21}N_3H_{15})$ ($X=\text{Ta, Hf}$) can be comprehended by the built TB model. The DFT calculations predict that the SOC can result in both of the large band-gap QAH state (about 30 meV) and topological trivial (NMI) state in the metal-organic framework. And the normal magnetic insulating state can be tuned to a QAH state by decreasing the SOC. Our findings exhibit strong SOC interactions may not always produce topological nontrivial states with large band gaps, but also induce a topological phase transition from topological nontrivial states to trivial states. These results are helpful in understanding of, searching for, and designing of topological quantum materials based on the multiple p and multiple d orbitals in 2D material systems.

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